dissertation under evaluation

Threat Detection in Network Traffic using Time Series Analysis

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Encrypted Network Traffic

The Standard for Privacy

Crucial for protecting user privacy, sensitive data, and critical infrastructure.

Encryption

The payload is encrypted by cryptographic protocols, ensuring that content remains confidential during transit.

Limited Visibility

Network operators see only remaining pieces of information, such as metadata, rather than the raw payload.

Design-Enabled

Encryption is enabled by design across operating systems, modern web browsers, and enterprise applications.

Adoption of Encryption in Network Traffic

2010s
HTTPS adoption driven by Google

2018
TLS 1.3 standardised

2019-2020
QUIC / HTTP/3 deployment

2022
~95% of web traffic encrypted (Google Transparency Report)

2023-2024
IoT, OT, and mobile networks increasingly enforce TLS by default

Encrypted Network Traffic: Pros vs. Cons

Pros

- Privacy protection of users
- Secure transmission of passwords and credentials
- In modern system is combined with MAC to prevent data tampering in transit

Cons

- Payload inspection (DPI) is no longer possible (without proxy)
- Attack traffic can not be recognized in legitimate traffic at the byte level
- Parental control bypass
- Blocklists and metadata-only analysis is possible
- Malware authors deliberately exploit encryption to evade detection

Threat Detection in Encrypted Network Traffic

Key Threat Categories Operating Over Encrypted Channels:

Malware C&C

Botnets (Mirai, Emotet) use HTTPS/DoH for command channels.

Cryptomining

Persistent, periodic beaconing to mining pools hidden in TLS streams.

APT Lateral Movement

Low-and-slow encrypted tunnels evade threshold-based detection.

Data Exfiltration

Covert channels within HTTPS, DNS, or QUIC.

Why is this hard?

- **No payload:** Signature-based IDS (Snort, Suricata) fail.
- **Valid Certificates:** Attackers easily obtain valid certs (e.g., Let's Encrypt).
- **Decryption is not possible:** Deploying intermediary proxy servers for traffic investigation is not possible everywhere

Analysis of Encrypted Network Traffic

Without payload access, analysis relies on observable metadata:



IP Flow Features

Duration, byte/packet counts, inter-arrival times, and flags.

Standardized via NetFlow or IPFIX protocols.



TLS Fingerprinting

JA3/JA4 hashes of ClientHello fields.

Challenge:
Partially effective but easily spoofed by advanced threats.

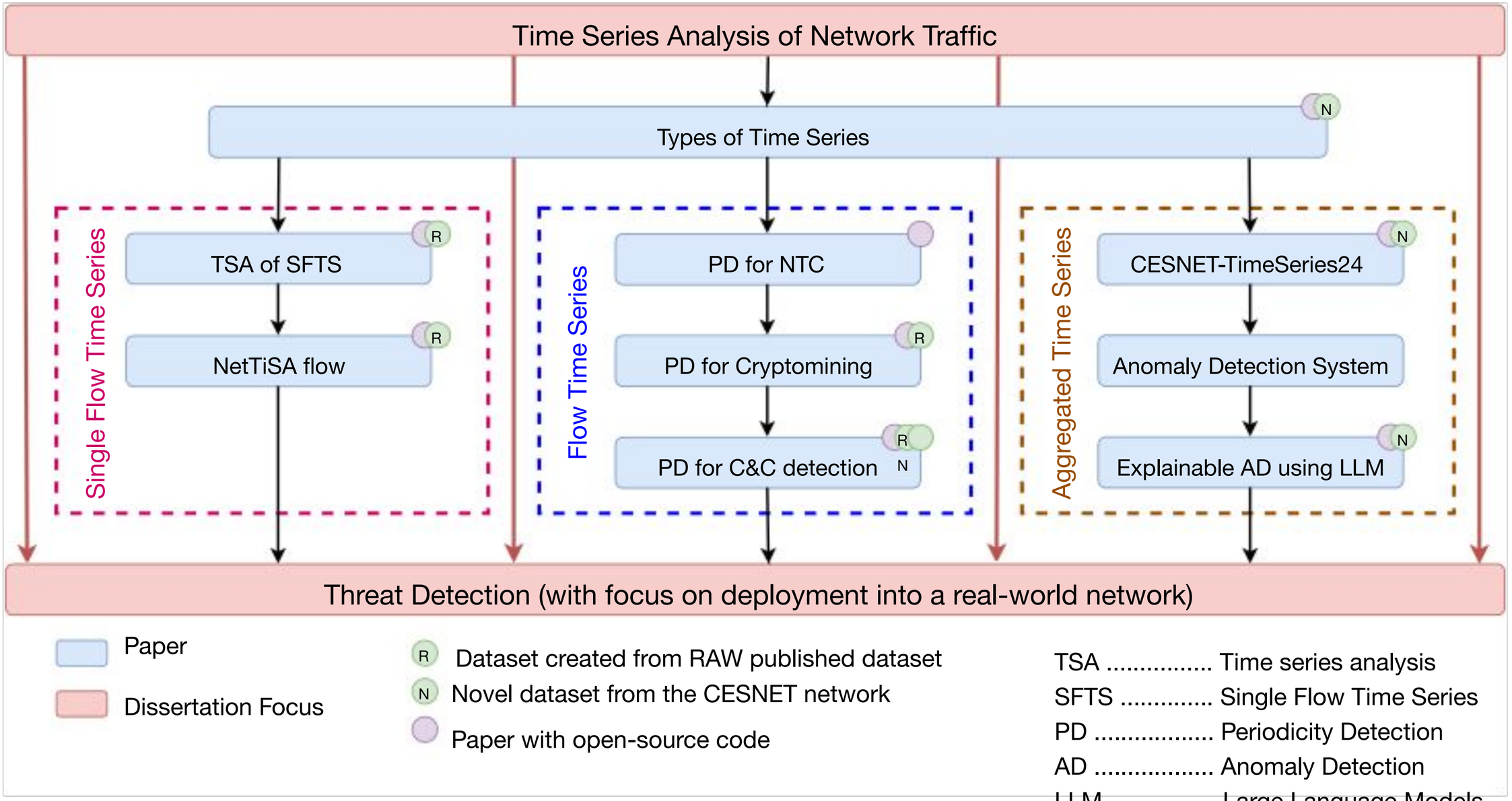


Certificate Metadata

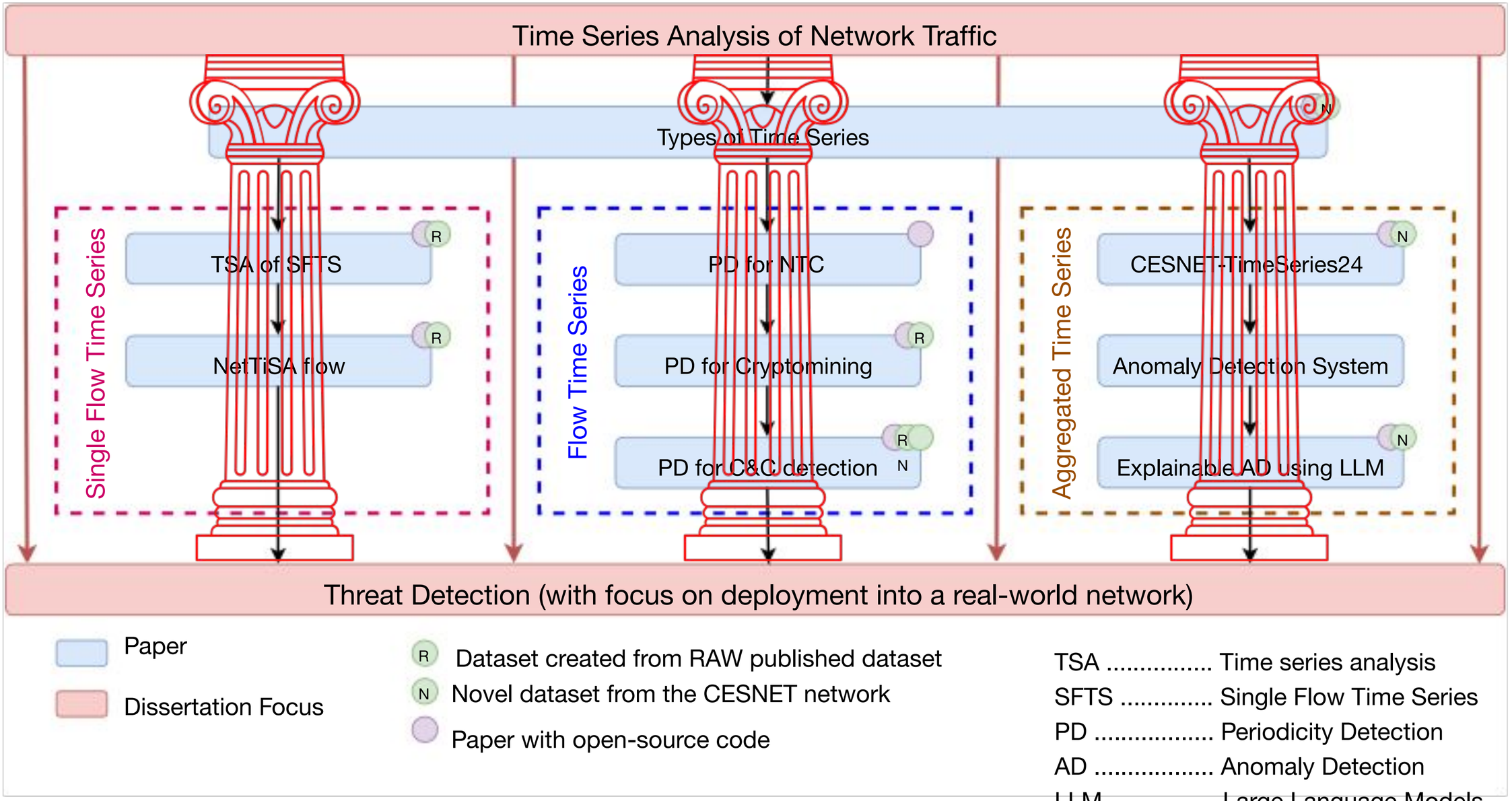
Subject, issuer, and validity period information.

Provides identity context without decrypting the payload.

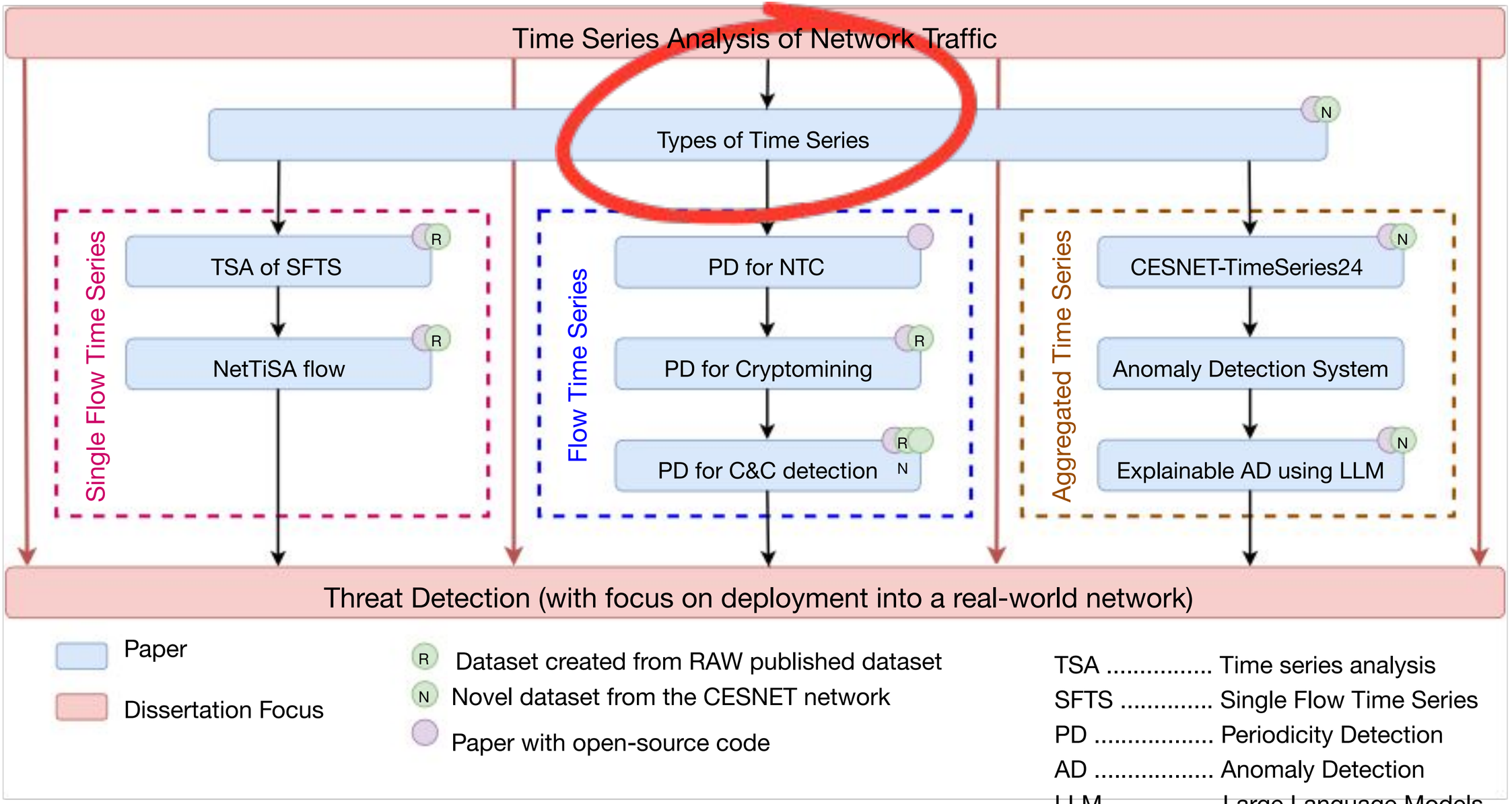
Overview of the Dissertation



Overview of the Dissertation



Time Series from Network Traffic



Time Series from Network Traffic

Packet TS

Data point: packet

Advantage: view without aggregation

Disadvantage: processing time, noise

Single Flow TS

Data point: packet

Advantage: without noise

Disadvantage: short time view

Flow TS

Data point: IP flow

Advantage: larger time view

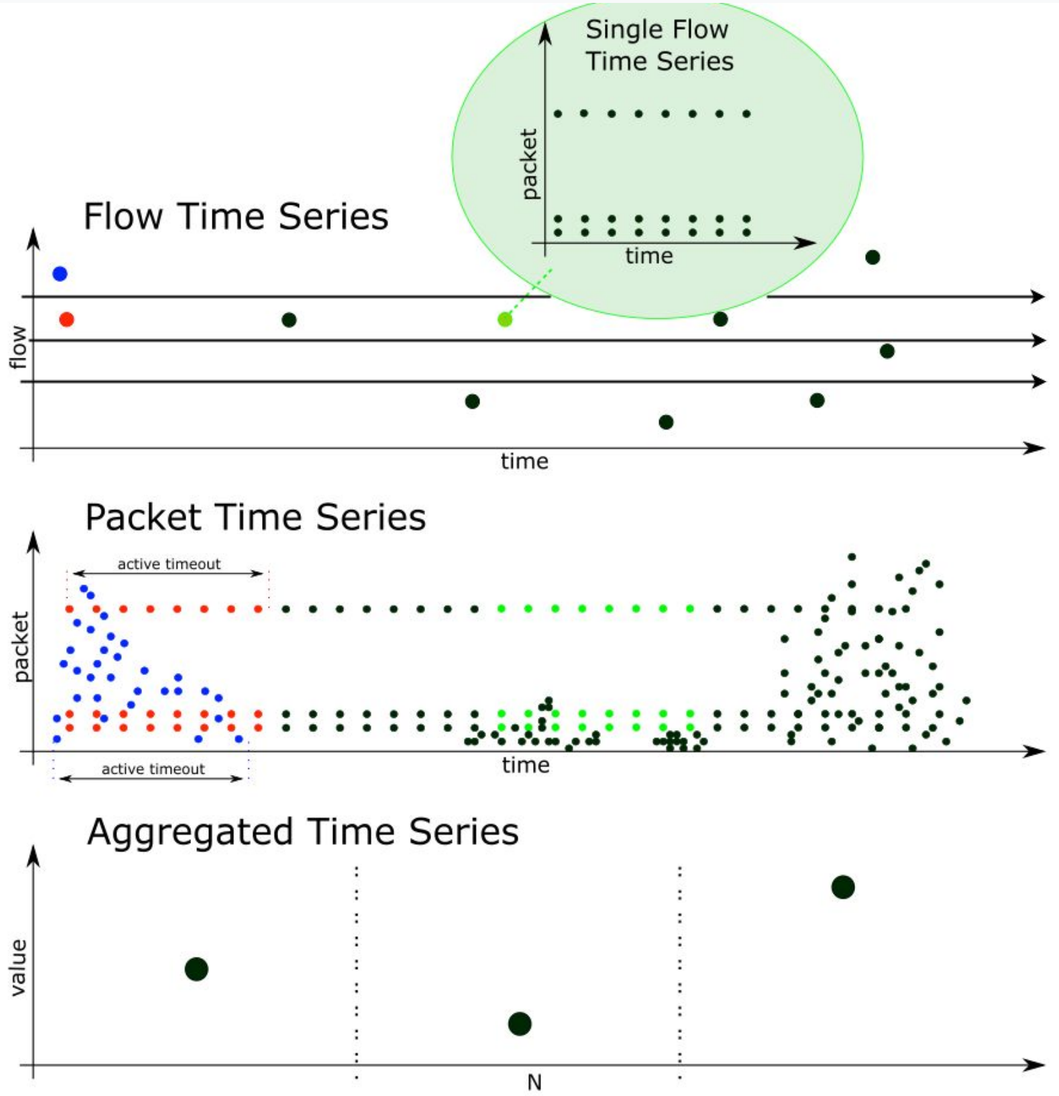
Disadvantage: aggregation

Aggregated TS

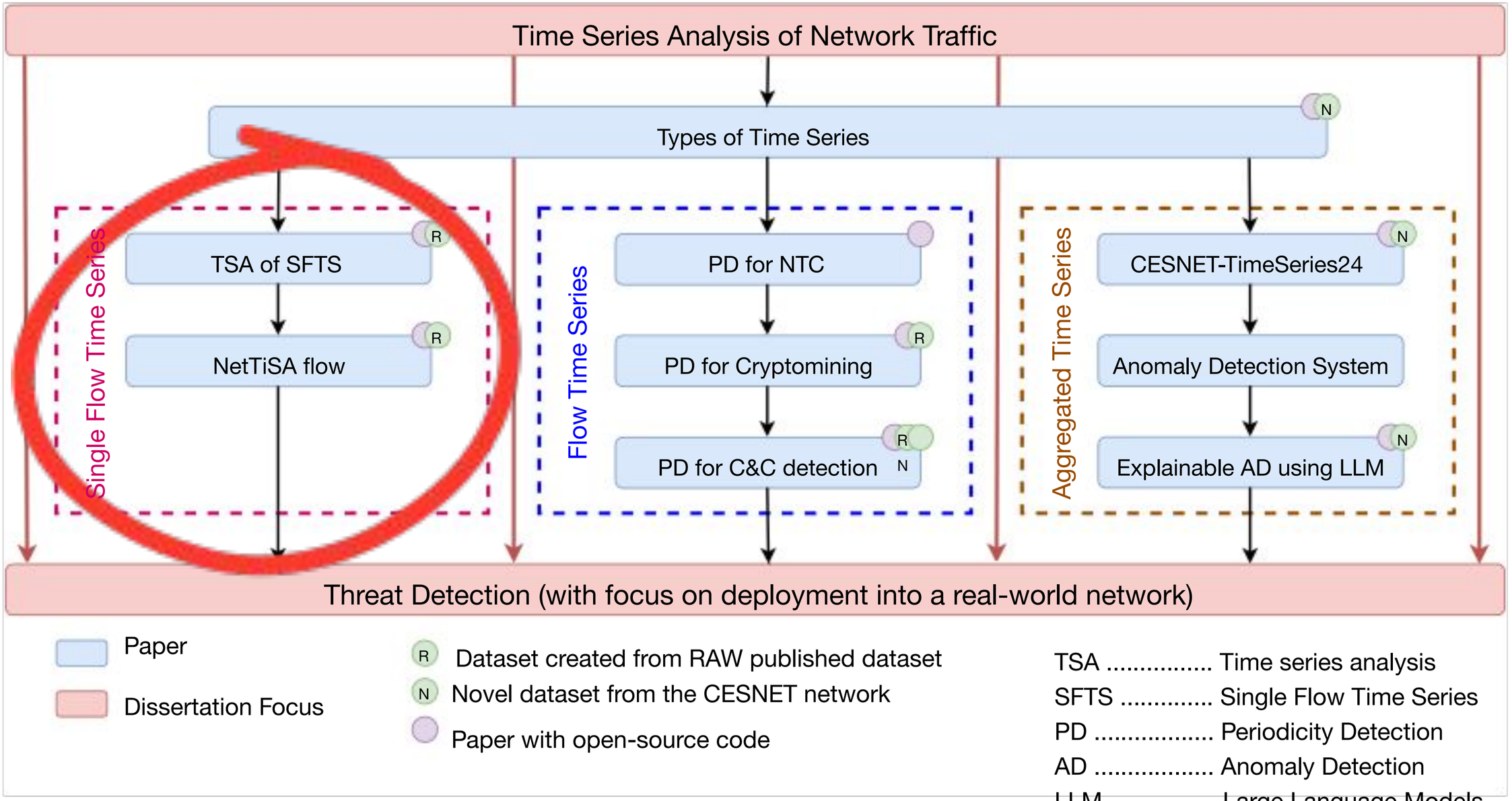
Data point: aggregated value

Advantage: long time view

Disadvantage: aggregation, missing data



Threat Detection using Single Flow Time Series Analysis



Unique Threat Behaviors in Single Flow Time Series

Key Observation

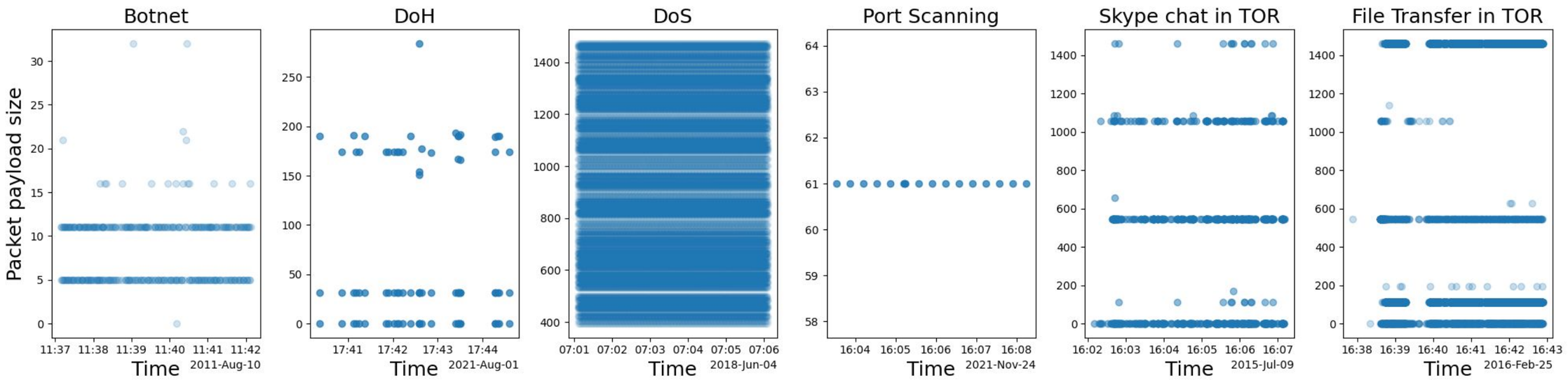
Threats exhibit distinct, identifiable behaviors within time series (packet sequences).

Verified by research using SPLT sequences and packet matrices.

Existing Limitations

- **Information Loss:** Exporting only the first N packets misses critical behavior.
- **Deployment Barriers:** Exporting it is computationally non-deployable in real-time.

Behavioral Fingerprints by Threat Type



Unique Threat Behaviors in Time Series

69 Extracted Features

Captured via Time Series Analysis:

- **Statistical-based:**
  Mean, Median, Std dev, ...
- **Time-based:**
  Relative times, Duration, ...
- **Frequency-based:**
  Power mean, Spectral flatness, ...
- **Distribution-based:**
  Hurst, Normal distribution, ...
- **Behavioral-based:**
  Transients, Periodicity, ...

Evaluation Tasks

Binary and Multiclass detection of:

- Botnet detection/classification
- Cryptomining detection
- DNS malware detection
- DNS over HTTPS detection
- DoS attack detection
- HTTPS Bruteforce detection
- Intrusion detection/classification
- IoT malware classification
- TOR detection/classification
- VPN traffic detection/classification

Our approach for threat detection (TSA of SFTS)

Traffic Detection (Binary)

- **Better** on 8 datasets
- **Similar** on 6 datasets
- **Worse** on 1 dataset

Traffic Classification (Multitask)

- **Better** on 5 datasets
- **Similar** on 1 dataset
- **Worse** on 2 datasets

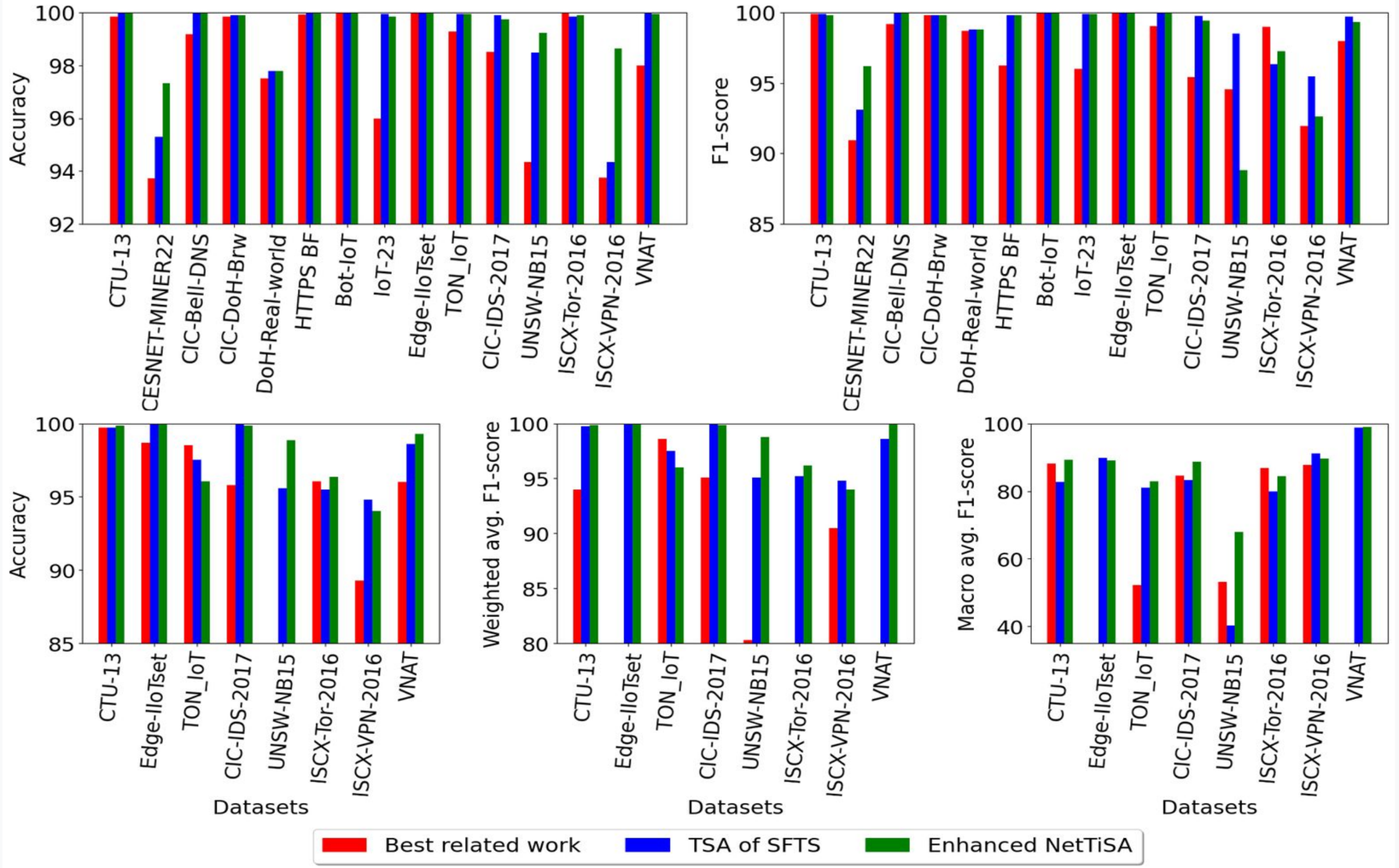
Averaged results compared to best-related work:

	Average	Accuracy	F1-score
Overall	Macro	-0.03 (± 0.70)	-0.08 (± 2.01)
Binary	—	-0.11 (± 0.29)	-0.09 (± 0.24)
Multiclass	Macro	0.02 (± 1.32)	0.06 (± 3.60)
	Weighed	0.02 (± 1.32)	-0.19 (± 0.99)

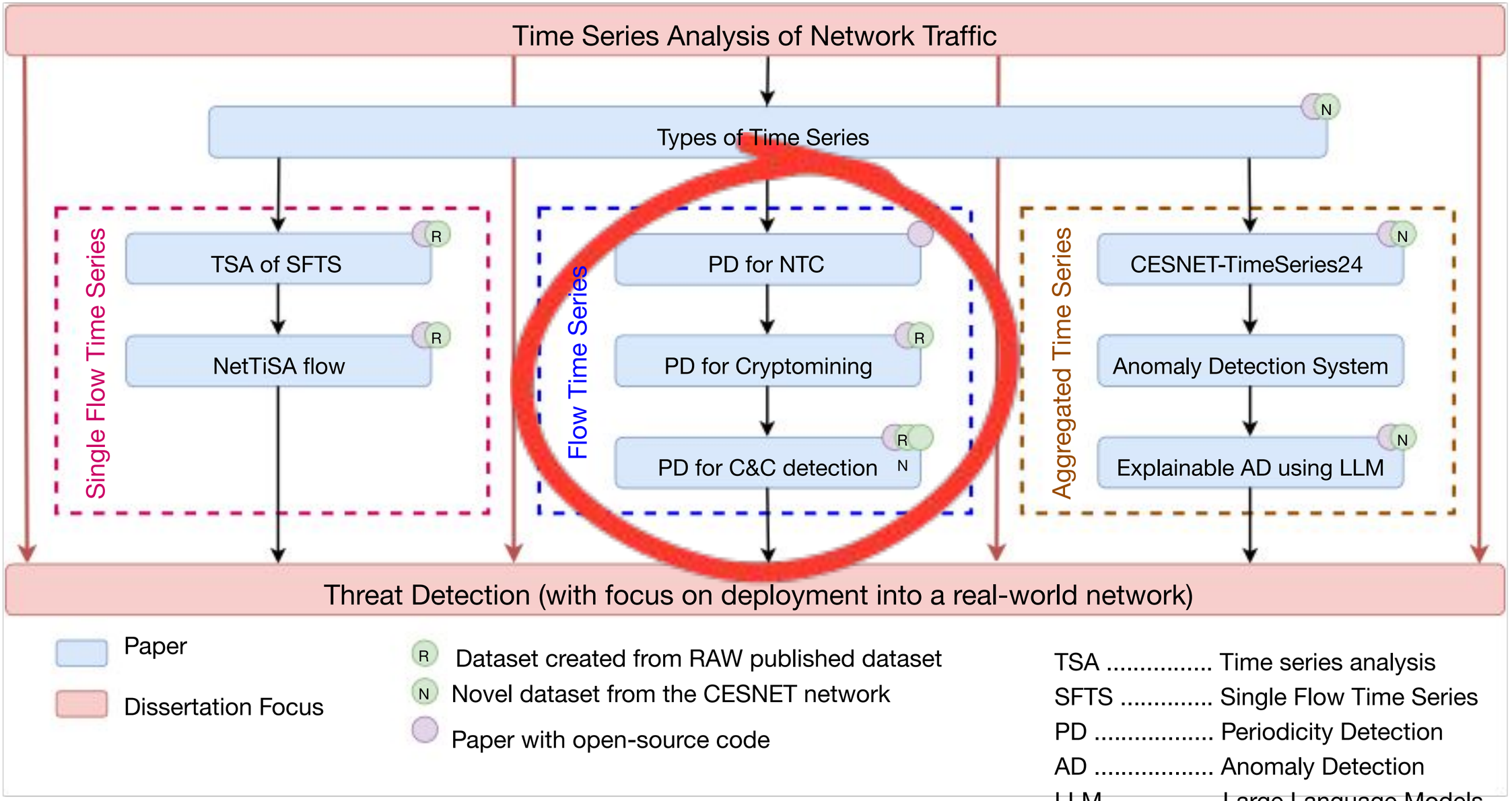
Our deployable approach (NetTiSA flow)

NetTiSA flow

The flow extension which can be computed in real-world ISP network with smallest increase of telemetry compared with similar approaches



Threat Detection using Flow Time Series Analysis

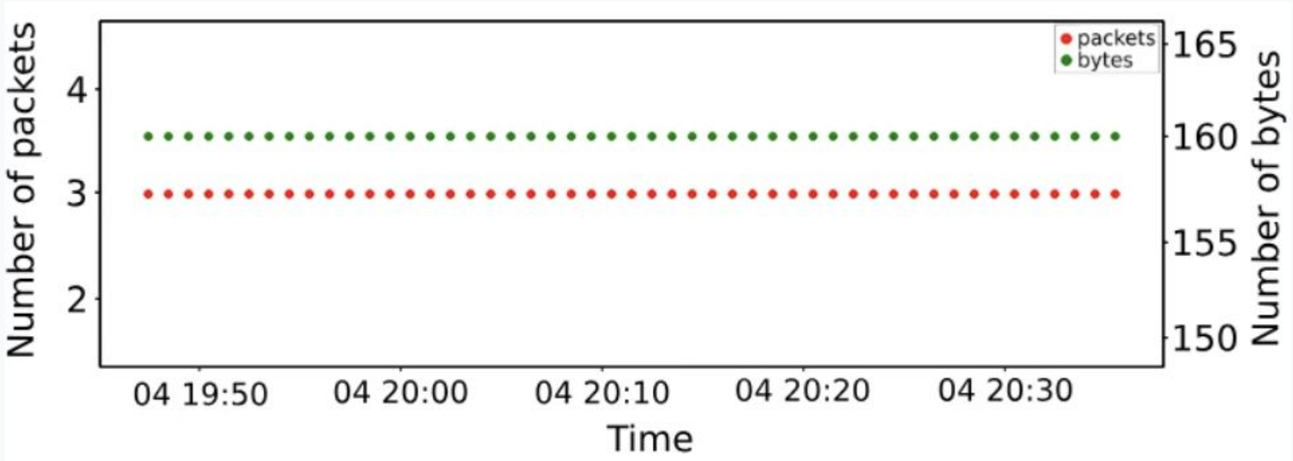


Flow time series usually contains significant periodicities

Two distinct types of periodic behavior are identified within the Flow time series:

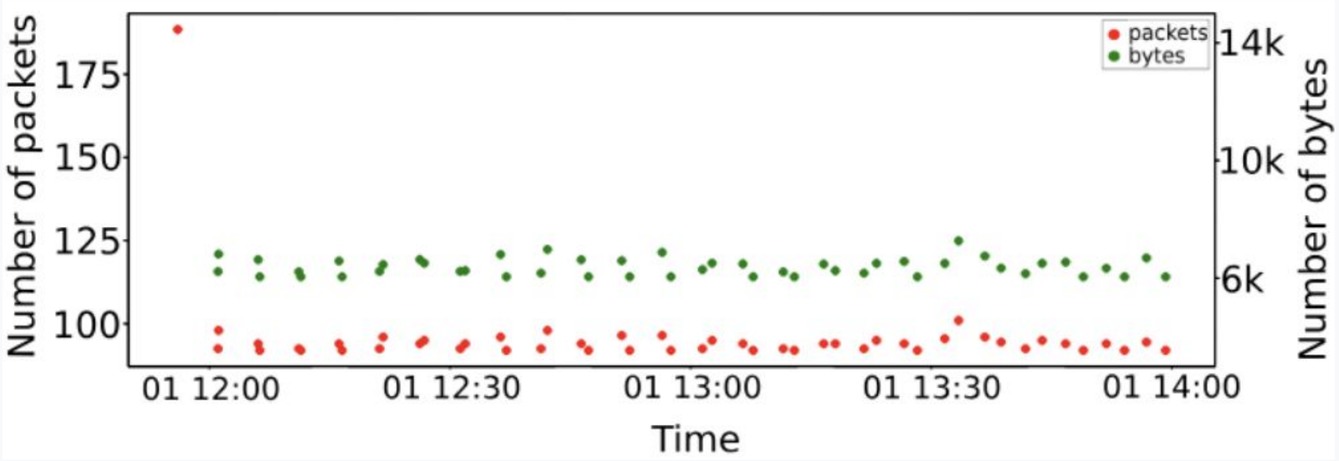
1. Clear

Repetition of constant values



2. Sinusoidal

Distributed periodically in time in “small” volume interval

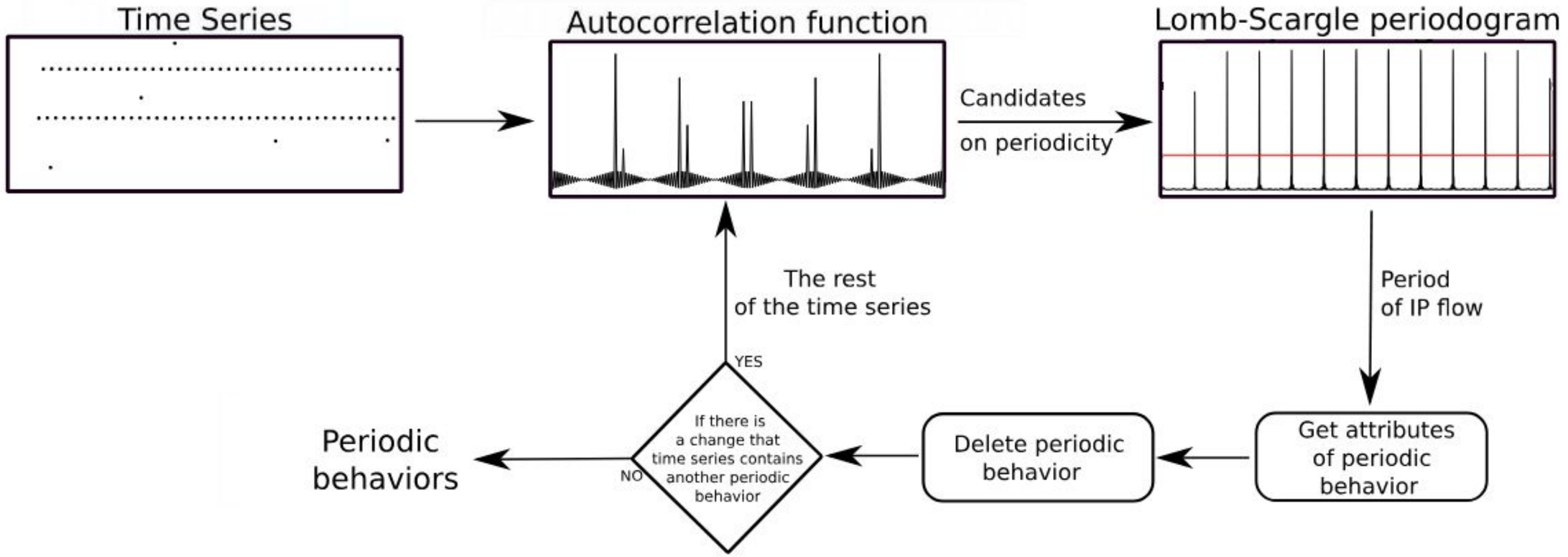


Detection of Periodic Behaviors

Periodicity detection by autocorrelation function and Lomb-Scargle periodogram

- Mainly for filtering non-periodic series

Attributes of periodic behaviors are used for **threat detection and classification**.



Detection and Classification Performance

Traffic classification

Classification of Applications, Services, and Operating Systems

91%
Macro F1-score

Cryptomining

Real-time detection of unauthorized mining activities

92%
Macro F1-score

Botnet C&C

Detection of Botnet's C&C Communication

100%
Precision

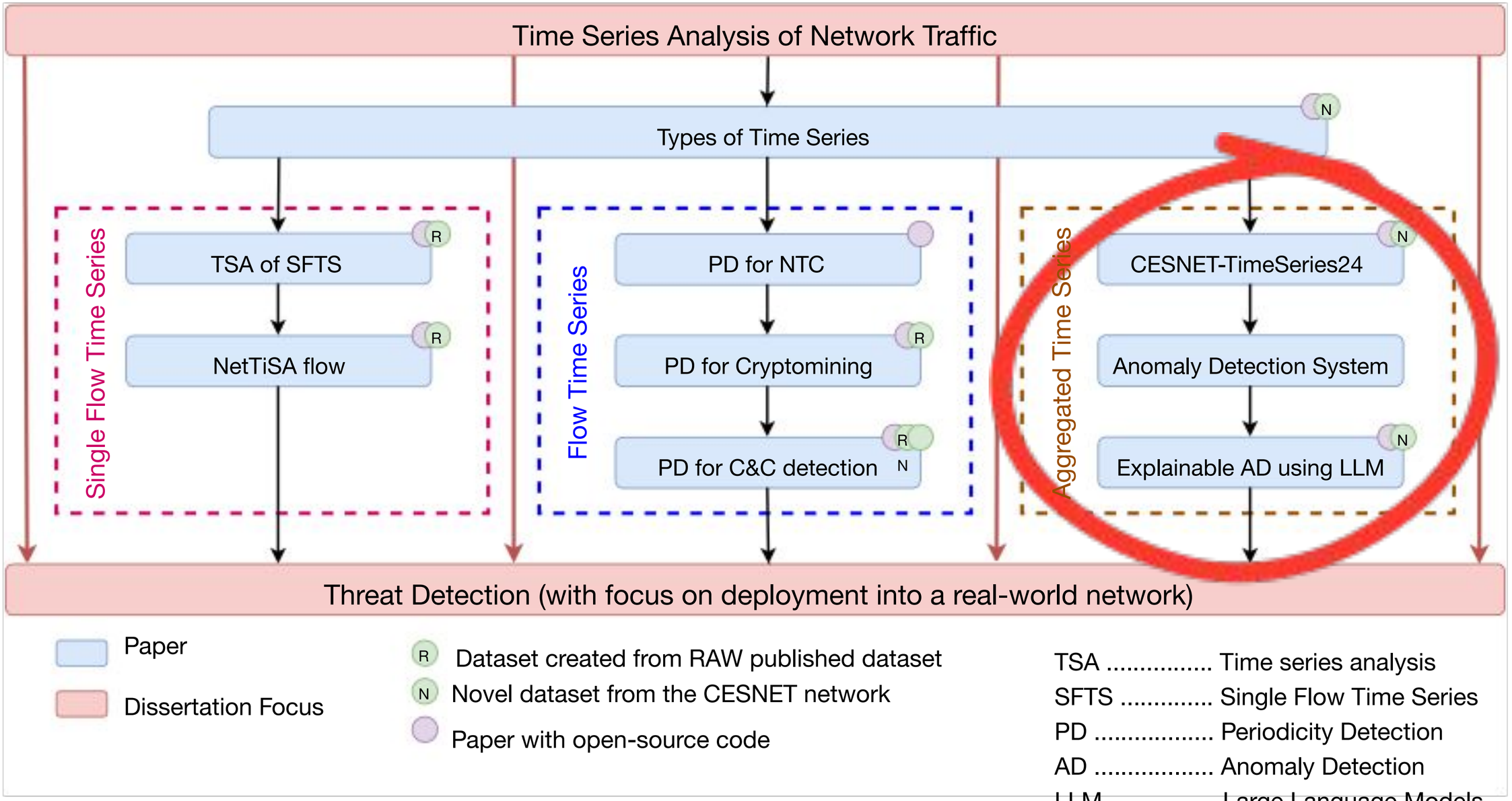
74.81
Macro F1-score

Machine Learning Methodology

All results achieved using XGBoost with rigorous evaluation protocols:

- Standard train / validation / test split
- Preprocessing fit on training data only
- Hyperparameters optimized on train and validation sets
- Final evaluation performed exactly once on test set

Threat Detection using Aggregated Time Series Analysis

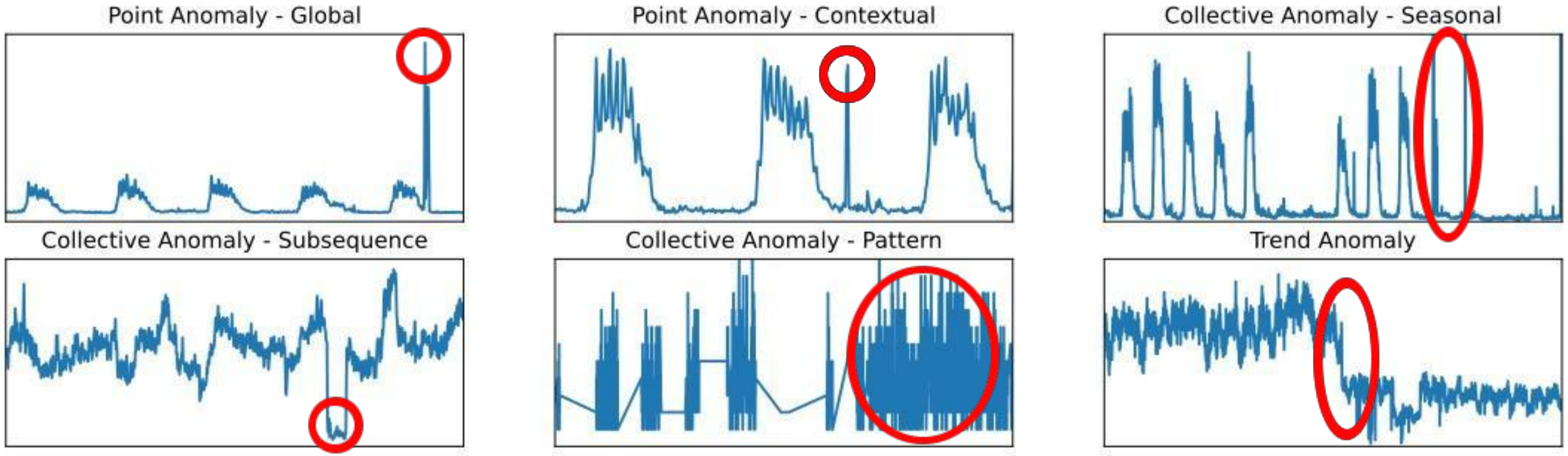


Anomalies in Network Traffic

Challenges with Supervised Classification

- Data drifts → models need to be retrained
- Network-specific setups → low transferability
- Target environment training required → difficult to obtain malware traffic

Unsupervised anomaly detection methods overcome these limitations



CESNET-TimeSeries24 Dataset

Time Series Metrics

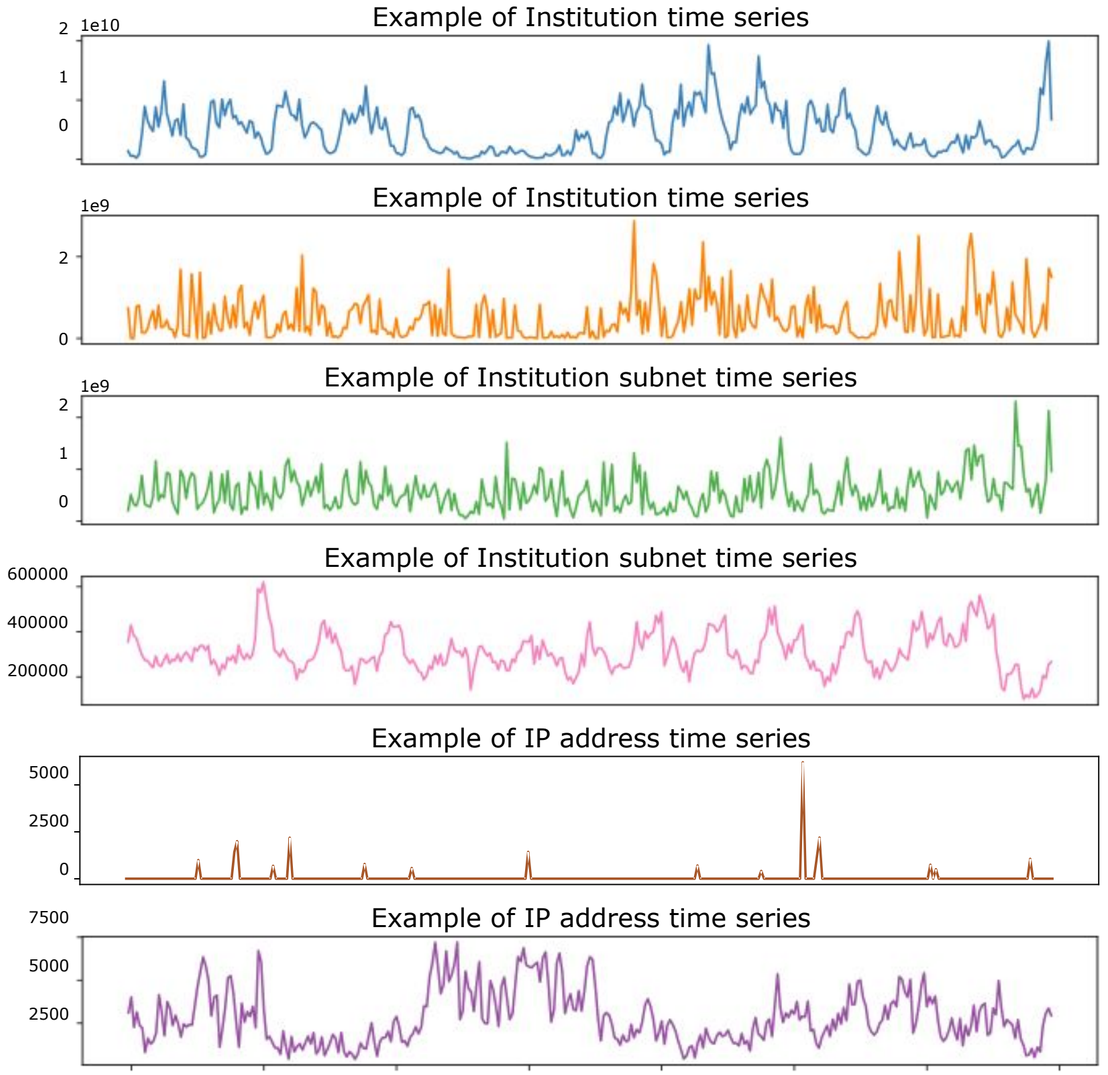
- IP flows, packets, and byte counts
- Unique destination IPs, ASNs, and countries
- TCP/UDP and packet direction ratios
- Average TTL and flow duration

Aggregations

10 minutes, 1 hour, 1 day

Identifiers Tracked

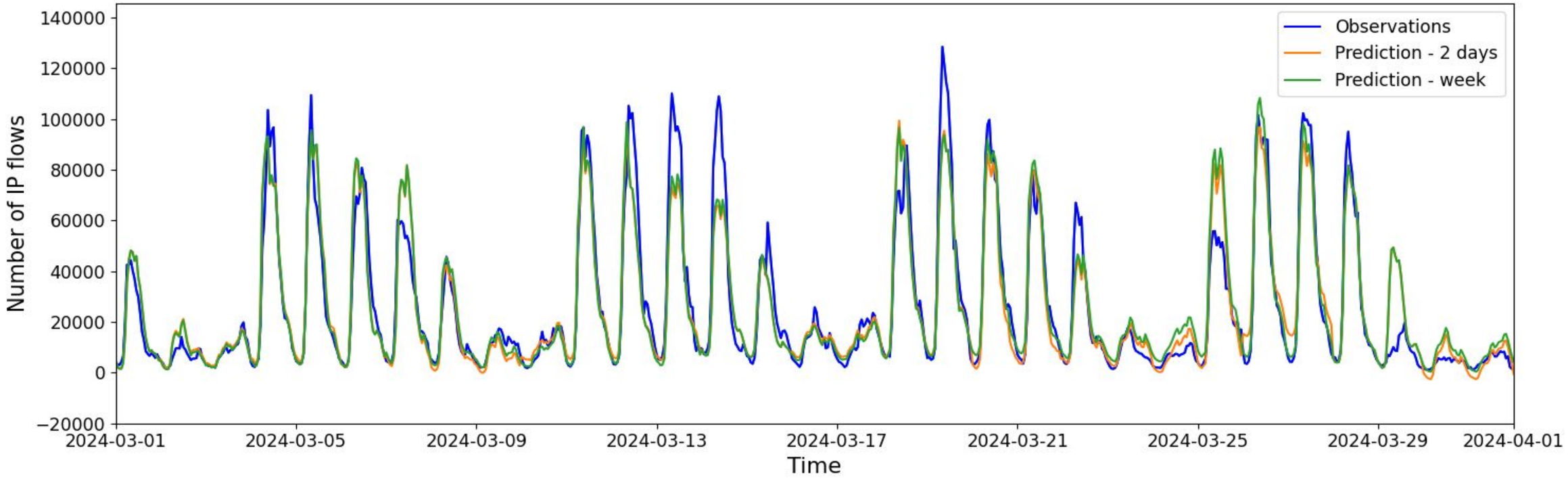
275,124 IP addresses
283 Institutions
548 Institution subnets



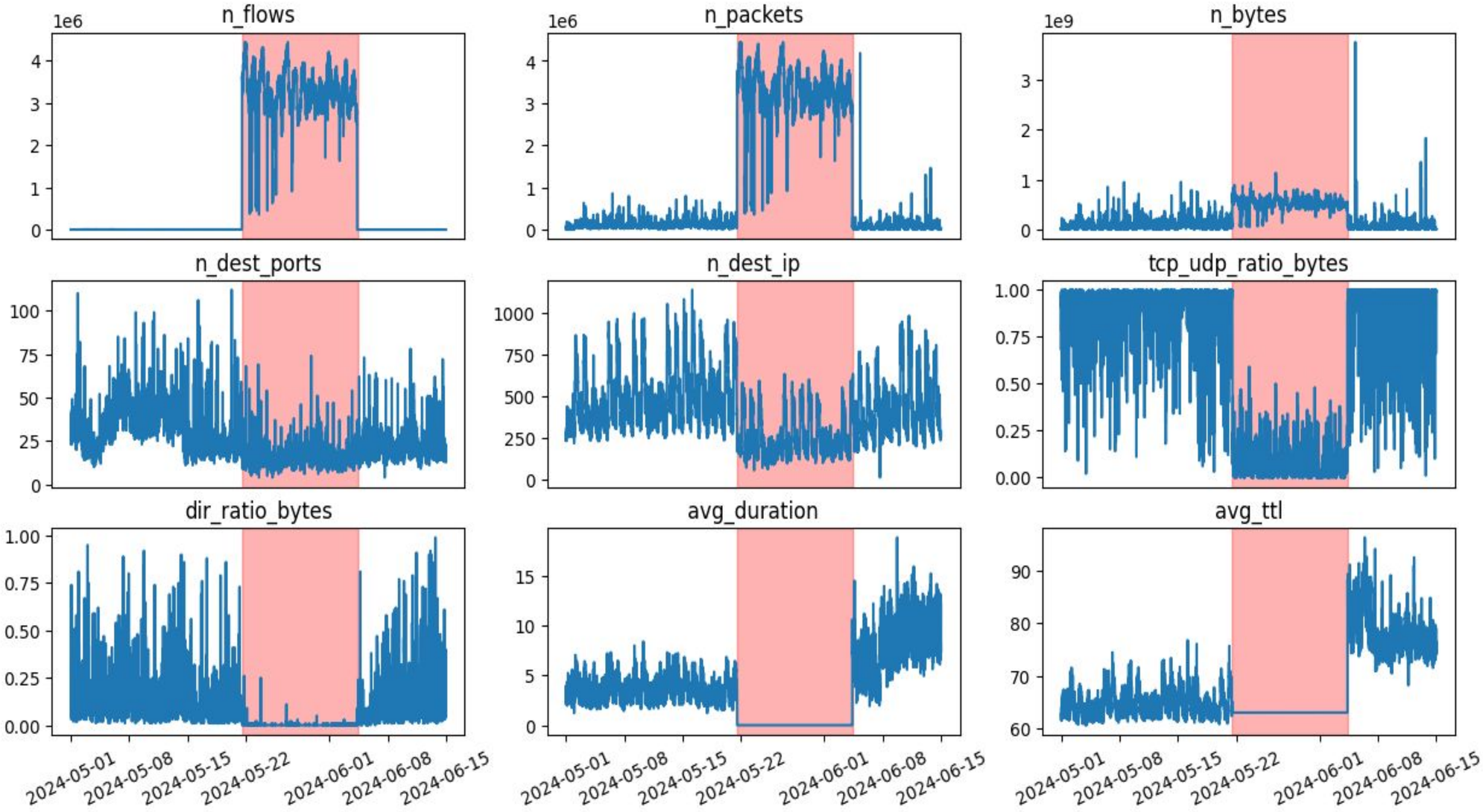
Anomaly Detection in Network Traffic

Possible unsupervised anomaly detection methods:

- Time series forecasting
- Distance-based methods (clustering and so on)
- Representation learning



Anomaly Detection in Network Traffic



Explainable Anomaly Detection

1. Traffic Analysis

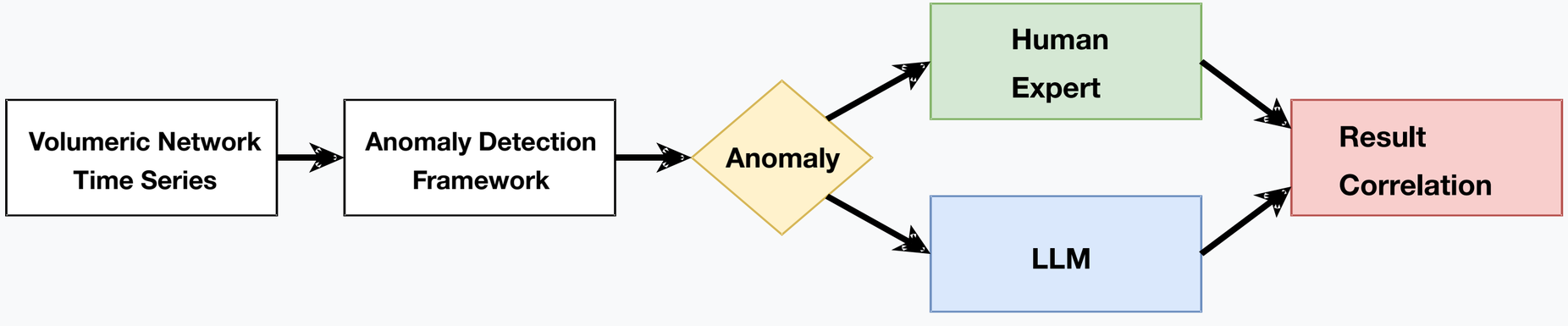
- Capture network traffic
- Create multivariate time series

2. Detection Framework

- Time series forecasting
- Compare forecast vs observed

3. Explainability

- Prepare anomaly context
- Prompt Large Language Model



Conclusion: Key findings

1. Feature Performance

Universal TSA feature set outperforms or matches state-of-the-art in binary/multiclass classification.

2. Periodic Behavior

Novel approach for apps, OS, cryptomining, and botnet C&C detection using periodic behavior.

3. Anomaly Detection

System utilizing forecasting-based methods proposed for robust anomaly detection.

4. LLM Explanability

LLMs successfully utilized for explaining detected anomalies in real-world network data.

5. Novel Datasets Created & Published

- **CESNET-TimeSeries24:** Anomaly detection and forecasting
- **CESNET-CC25:** Packet, flow, and time series for C&C
- **CESNET-MINER22-TS:** Time series for cryptomining detection
- **CESNET-USTS23:** Statistical and behavioral analysis
- **CESNET-DeviceType24:** Device type classification

Datasets:

Conclusion: The Impact of my Research

Publications

- **16 peer-reviewed** articles on encrypted traffic analysis
- Directly relevant to the **thesis topic**

Open Science

- **9 datasets** published
- Consistent **source code release** for transparency
- All published in accordance with ethics statement approved by Ethical Committee of CTU in Prague

Thesis Scope

- **9 research papers** included in dissertation
- **Two journals** (Computer Networks, Nature Scientific Data)
- **Core B** conferences

"I have consistently released the source code and all supporting materials for every publication to ensure full transparency and reproducibility of my research."

Thank you for your attention

Josef Koumar

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Dissertation thesis: